

Table 1: The inference time comparison on Weather dataset, “TA” represents text-attribution.

Method	TADIFF	TADIFF w/o TA	VerbalTS	Sundial
Inference time (ms)	340	183	463	272

Table 2: The retrieval accuracy of different DTTC model on Synth dataset. Given a future sequence, DTTC-I and DTTC-E report the accuracy of retrieving the most similar historical sequence and future condition, respectively, from three candidates.

Setting	Original text encoder + Contrastive loss	Long-CLIP text encoder + Contrastive loss	Original text encoder + Cross-entropy loss
DTTC-I	84.57%	87.21%	82.71%
DTTC-E	96.36%	97.14%	96.71%

Table 3: Forecasting performance on Synth dataset, evaluated by different DTTC models with different encoder architectures and different optimization targets. The best results are in **bold**, and the second-best results are underlined. Arrow $\uparrow$ indicates that higher is better.

	Metric	DLinear	PatchTST	Sundial	VerbalTS	TimeCMA	TimeMMD	CT	IATSF	TADIFF (ours)
LongClip text encoder + Contrastive loss	$\uparrow$ DTTC-I	19.37 $\pm$ 0.34	<u>20.47</u> $\pm$ 0.04	19.82 $\pm$ 0.02	17.26 $\pm$ 0.02	19.12 $\pm$ 0.04	20.02 $\pm$ 0.08	19.49 $\pm$ 0.01	18.23 $\pm$ 0.04	21.59 $\pm$ 0.10
	$\uparrow$ DTTC-E	31.15 $\pm$ 0.29	<u>33.70</u> $\pm$ 0.05	30.59 $\pm$ 0.01	<u>41.99</u> $\pm$ 0.19	30.80 $\pm$ 0.07	35.45 $\pm$ 0.20	34.79 $\pm$ 0.22	29.87 $\pm$ 0.02	44.33 $\pm$ 0.04
Original text encoder + Cross-entropy loss	$\uparrow$ DTTC-I	13.29 $\pm$ 0.15	<u>14.21</u> $\pm$ 0.04	13.28 $\pm$ 0.01	12.12 $\pm$ 0.04	13.22 $\pm$ 0.00	13.88 $\pm$ 0.04	13.05 $\pm$ 0.05	12.65 $\pm$ 0.03	15.22 $\pm$ 0.04
	$\uparrow$ DTTC-E	72.59 $\pm$ 0.18	<u>75.54</u> $\pm$ 0.06	73.02 $\pm$ 0.03	<u>83.22</u> $\pm$ 0.24	72.23 $\pm$ 0.07	77.32 $\pm$ 0.22	76.23 $\pm$ 0.28	71.34 $\pm$ 0.04	85.20 $\pm$ 0.11

Table 4: Counterfactual forecasting performance on the Weather dataset. We only evaluate the samples with entirely new types of events that never occur in the train split, which include 674 samples. The best results are in **bold**, and the second-best results are underlined. Arrow $\uparrow$ indicates that higher is better.

Metric	DLinear	PatchTST	Sundial	VerbalTS	TimeCMA	TimeMMD	CT	IATSF	TADIFF (ours)
$\uparrow$ DTTC-I	26.88 $\pm$ 0.03	26.87 $\pm$ 0.02	26.89 $\pm$ 0.02	24.47 $\pm$ 0.14	<u>26.90</u> $\pm$ 0.01	26.72 $\pm$ 0.11	26.21 $\pm$ 0.36	26.68 $\pm$ 0.01	26.99 $\pm$ 0.06
$\uparrow$ DTTC-E	28.30 $\pm$ 0.02	28.62 $\pm$ 0.02	28.24 $\pm$ 0.02	<u>29.81</u> $\pm$ 0.06	28.26 $\pm$ 0.01	28.35 $\pm$ 0.28	29.31 $\pm$ 0.32	28.46 $\pm$ 0.04	30.03 $\pm$ 0.07

Table 5: Factual and counterfactual forecasting with LLM revised text on Traffic dataset. The best results are in **bold**. Arrows $\uparrow$ ($\downarrow$) indicate that higher (lower) is better.

Setting	Metric	Sundial	TimeMMD	TADIFF (ours)
Factual	$\downarrow$ MAE	0.40 $\pm$ 0.00	0.43 $\pm$ 0.01	0.31 $\pm$ 0.00
	$\downarrow$ MSE	0.34 $\pm$ 0.00	0.29 $\pm$ 0.01	0.23 $\pm$ 0.00
	$\uparrow$ DTTC-I	21.97 $\pm$ 0.00	18.81 $\pm$ 0.21	22.90 $\pm$ 0.05
	$\uparrow$ DTTC-E	50.70 $\pm$ 0.04	56.41 $\pm$ 0.43	66.30 $\pm$ 0.15
Counterfactual	$\uparrow$ DTTC-I	21.94 $\pm$ 0.01	18.71 $\pm$ 0.32	22.87 $\pm$ 0.07
	$\uparrow$ DTTC-E	50.63 $\pm$ 0.01	53.46 $\pm$ 0.57	64.20 $\pm$ 0.05

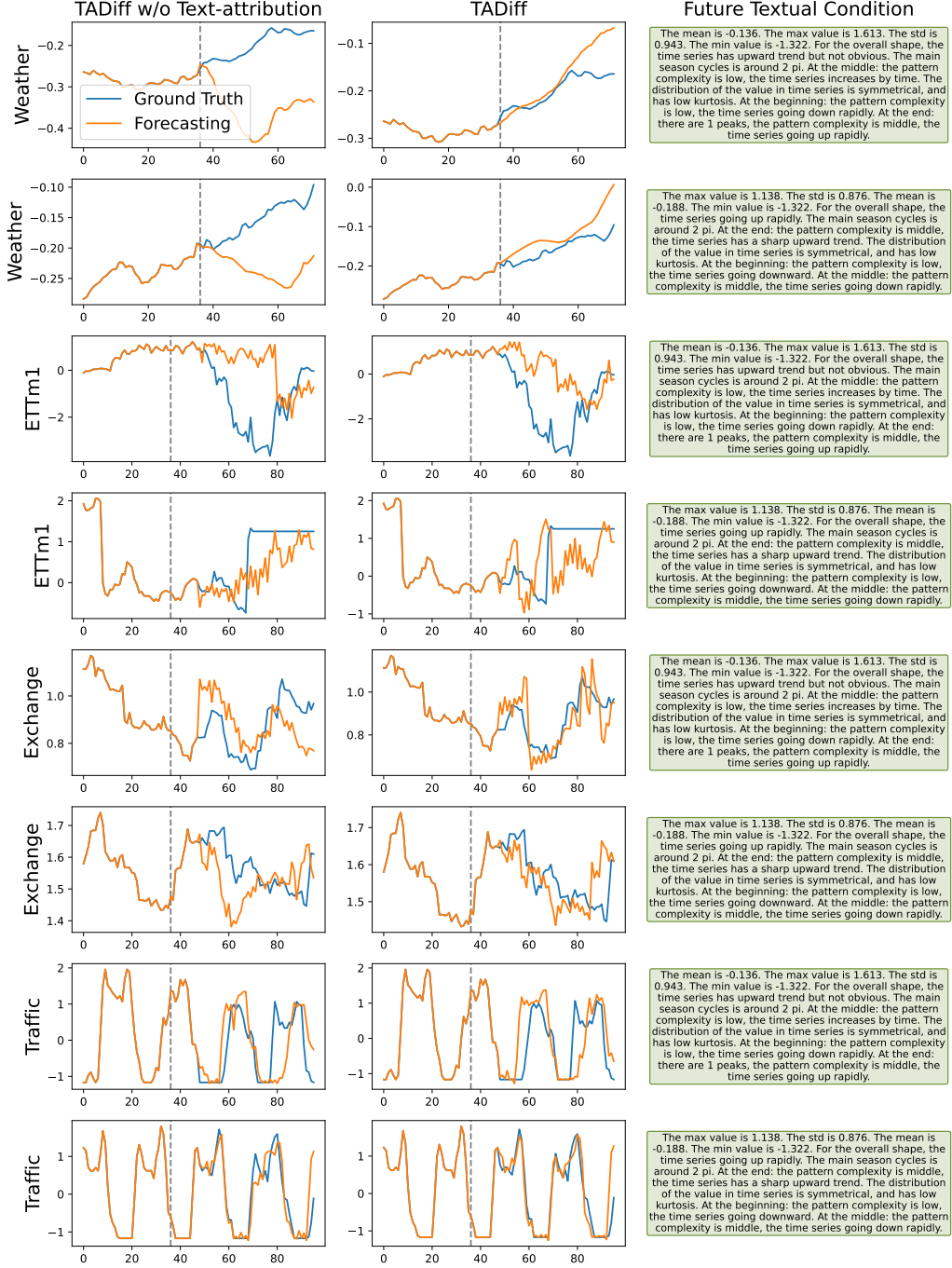


Figure 1: Qualitative comparison of forecasting on Weather, ETTm1, Exchange, and Traffic datasets. Column 1: the forecasting of TADiff without text-attribution; Column 2: the forecasting of TADiff with text-attribution; Column 3: the future textual condition.

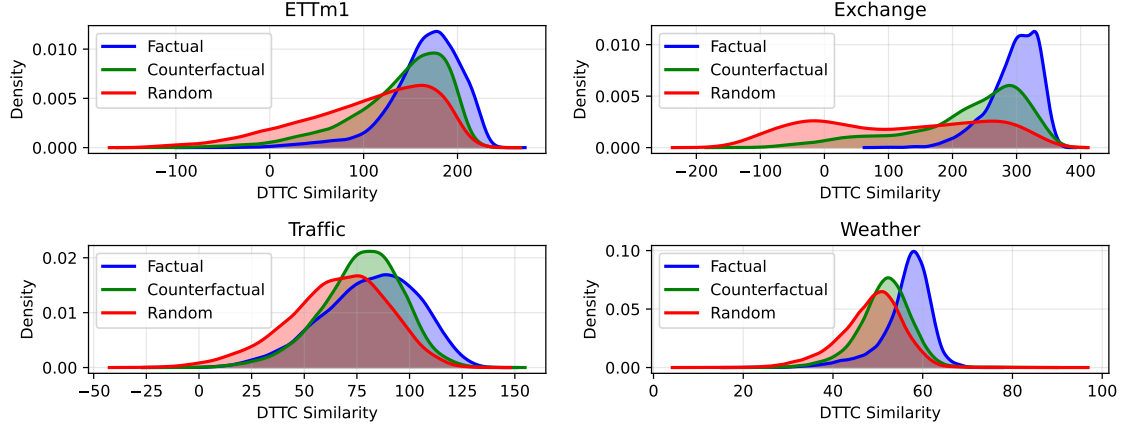


Figure 2: DTTC similarity across factual data, counterfactual data, and random data on four real-world datasets. “Factual” represents the factual data, “Counterfactual” represents the counterfactual data constructed by us, “Random” represents the counterfactual data constructed through totally random sampling.

Templated Text

The distribution of the value in time series is symmetrical, and has low kurtosis.
 For the overall shape, the time series decreases by time.
 The main season cycles is around 2π .
 At the end: the time series has a sharp downward trend.
 At the middle: the time series going up rapidly.
 At the beginning: there are 1 peaks, the time series has a slow upward trend.



LLM-revised Text

The distribution of values over time is symmetric and exhibits low kurtosis. Overall, the series shows a decreasing trend. The primary seasonal cycle corresponds to a period of approximately 2π . At the beginning, the series features one peak with a gradual upward trend. In the middle segment, the series rises sharply. Toward the end, the series undergoes a pronounced downward trend.

Figure 3: An example of the differences between templated-style texts and LLM-revised texts on the Traffic dataset.